EX No:1a	SETTING UP PYTHON ENVIRONMENT AND LIBRARIES – JUPYTER NOTEBOOK

Aim:

To install Python, configure Jupyter Notebook, understand the notebook environment, execute Python programs, create Markdown cells, and use basic Jupyter Notebook features for Data Analytics applications.

Algorithm:

Algorithm 1: Installing Python

1. Download Python from python.org
2. Run the installer
3. Select "Add Python to PATH"
4. Click Install Now
5. Verify installation using: python --version
6. Observe installed Python version

Algorithm 2: Installing Jupyter Notebook

1. Open Command Prompt
2. Install Jupyter pip install notebook
3. Launch Jupyter Notebook jupyter notebook
4. Browser automatically opens Notebook Dashboard.

Algorithm 3: Creating a Notebook

1. Open Jupyter Dashboard
2. Select New → Python 3
3. Create Notebook
4. Rename Notebook
5. Create Code and Markdown Cells
6. Execute cells using Shift + Enter

Program:

Program 1 — Simple Python Program

```
print("Welcome to Data Analytics Lab")
```

Output:

```
Welcome to Data Analytics Lab
```

Program 2 — Arithmetic Operations

```
a = 20
b = 10
print("Addition =", a + b)
print("Subtraction =", a - b)
print("Multiplication =", a * b)
print("Division =", a / b)
```

Output:

```
Addition = 30
Subtraction = 10
Multiplication = 200
Division = 2.0
```

Program 3 — Display Student Details

```
name = "John"
dept = "AI&DS"
cgpa = 8.9
print("Name:", name)
print("Department:", dept)
print("CGPA:", cgpa)
```

Output:

```
Name: John
Department: AI&DS
CGPA: 8.9
```

Program 4 — Creating Markdown Cell

```
# Data Analytics Lab
## Experiment 1
### Jupyter Notebook
```

Then press **Shift + Enter**.

Program 5 — Using NumPy Library

```
import numpy as np
a = np.array([10, 20, 30, 40, 50])
print(a)
print("Mean =", np.mean(a))
print("Sum =", np.sum(a))
```

Output:

```
[10 20 30 40 50]
Mean = 30.0
Sum = 150
```

Program 6 — Using Pandas Library

```
import pandas as pd
data = {
    'Name': ['John', 'David', 'Alex'],
    'Marks': [85, 90, 95]
}
df = pd.DataFrame(data)
print(df)
```

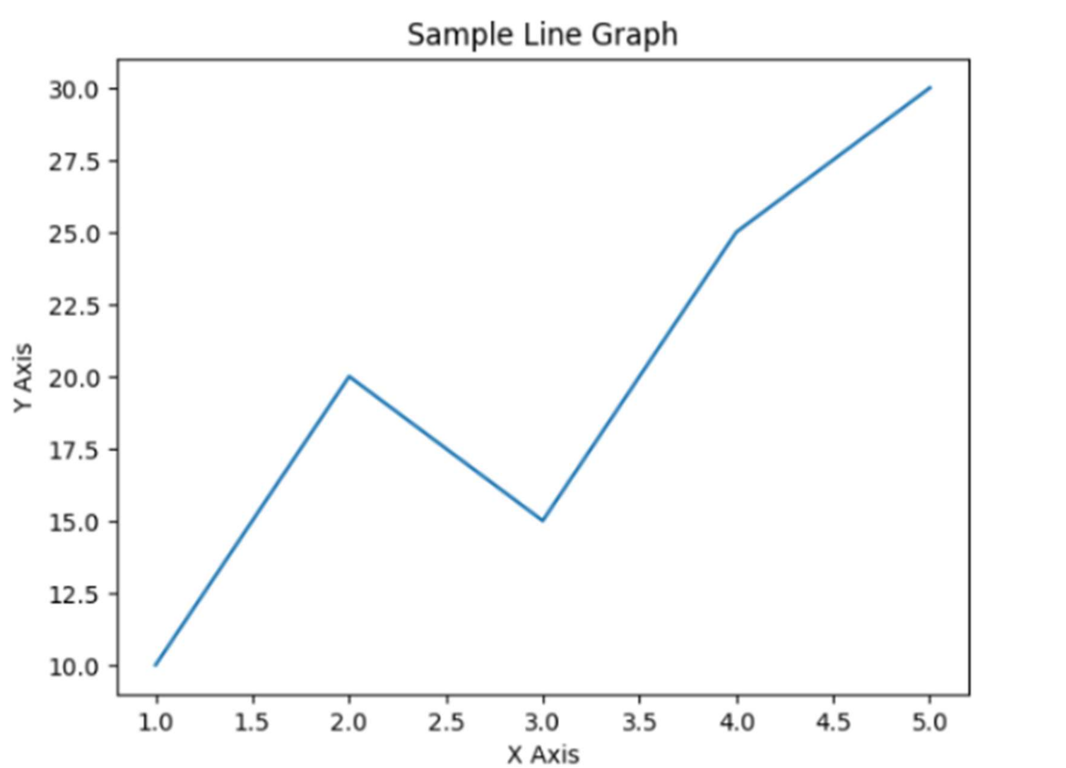
Output:

```
    Name  Marks
0   John     85
1  David     90
2   Alex     95
```

Program 7 — Data Visualization

```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y = [10, 20, 15, 25, 30]
plt.plot(x, y)
plt.title("Sample Line Graph")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```

Output:



Result: